Name _____
Student Number _____

Tutorial 2 Worksheet

I- Add a checkmark or an X in the following table address the definitions of permissive and susceptible and describe why/why a permissive infection will occur.

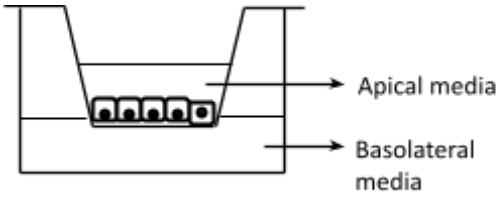
	Has Receptor?	Supports Replication?	Can a Productive Infection Occur?
Permissive ONLY		✔	
Susceptible ONLY	✔		
Susceptible and Permissive	✔	✔	✔

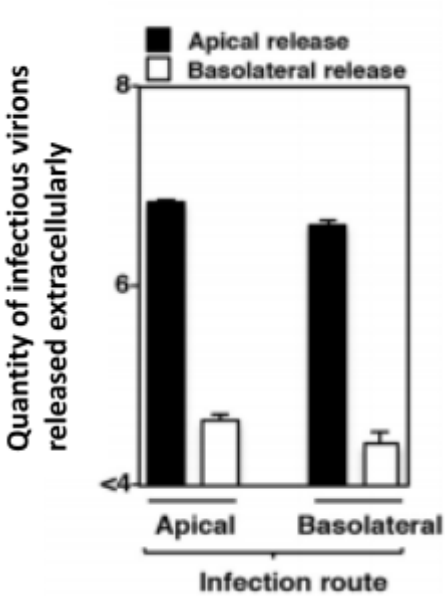
II. If the genome ONLY is inserted into a permissive host – explain if new virions will be able to be released.

	Describe the Mechanism	Will new virions be created (Y/N)
(+) sense RNA virus	The +RNA can be used as template directly, and it can also translate for RdRp. Therefore, it can just use the host ribosomes to produce its virions	Y
(-) sense RNA virus	The -RNA can not be used as template directly, therefore it needs to carry its own RdRp.	N

II. Analyzing experimental design and data (Past exam question)

Mumps virus (MuV) is an enveloped virus with a (-) sense RNA genome within the family *Paramyxoviridae*, and is the causative agent of mumps. MuV is transmitted through aerosols or by direct contact with contaminated respiratory secretions. Following exposure, the virus initially infects the upper respiratory tract epithelia and replicates in the cytoplasm. Katoh *et al.* tested how MuV infects polarized epithelial cells, and the direction in which new virions are released. Polarized respiratory epithelial cells were cultured such that the culture media on the top of the cells (apical media) could not mix with the media facing the bottom of the cells (basolateral media) as shown in the diagram. Cells were infected via two different infection routes: either on the apical (top) or basolateral (bottom) surface. After infection, the media was collected from both surfaces and the amount of new virons released extracellularly was quantified. The results are shown in the graph.





A. What does it mean for a cell to be polarized? (1 mark)

Being polarized means that a cell has two surfaces that differ in structure and function.

B. Describe what the data shows. In comparing the quantity of infectious virions released into the extracellular media following infection, how do the two infection routes compare? (3 marks)

Both infection routes made similar amount of virus.
Most virions were released from the apical side,
and only a small amount from basolateral side.

C. What does this data tell you about whether MuV favours attachment and host cell entry on a particular membrane surface? Explain your reasoning. (2 marks)

MuV does not seem to strongly prefer one side for entry.
Infection from both apical and basolateral side can produce virus.

D. Host cells in the respiratory epithelium bind to glycoproteins embedded in the MuV envelope. The researchers were able to fluorescently label the host cells' MuV receptors. Considering the architecture of the upper respiratory epithelium, where on the host cell plasma membrane would you expect these fluorescent labels to be? Why? (2 marks)

I expect the receptors to be on both apical and basolateral membrane,
because virus can infect from both sides.

E. New MuV virions are assembled at the host cell plasma membrane. Based on this data, which region/surface of the plasma membrane would be assembling the majority of new virions? (1 mark)

Most new virions are probably assembled at the apical membrane,
because most virus are released from there.

F. How does the direction of viral release typically relate to whether an infection is systemic or local? It turns out MuV is a systemic infection. From the data presented here, does this surprise you? Can you propose two plausible mechanisms that could explain the ability of MuV to spread systemically? (3 marks)

Usually basolateral release can help virus spread into the body,
while apical release is more local. So it is a little surprising
because MuV mostly release apically. Maybe the small amount
of basolateral virus is enough to spread, or infected cells can
help carry the virus to other tissues.